Name: _____

WORKSHEET

Exponential functions

1 Which of the following are exponential functions? (Circle)

$f(x) = x^2$ $\quad y = 2^x$ $\quad y = \sqrt[4]{x^3}$ $\quad f(a) = \dfrac{3}{a^2}$ $\quad f(x) = e^x$

$y = \dfrac{x^5}{x-1}$ $\quad y = ab^{\lambda x}$ $\quad f(x) = 8x^n$ $\quad f(n) = 1.05x^n$ $\quad f(x) = 10x^2$

$y = \dfrac{3}{10^x}$

2 Complete the table given the following functions

a $f(x) = 2^x$

x	-3	-2	-1	0	1	2	3
$f(x)$							

b $f(x) = -2^x$

x	-3	-2	-1	0	1	2	3
$f(x)$							

c $f(x) = (-2)^x$

x	-3	-2	-1	0	1	2	3
$f(x)$							

d $f(x) = 10^x$

x	-3	-2	-1	0	1	2	3
$f(x)$							

e $f(x) = 10^{-x}$

x	-3	-2	-1	0	1	2	3
$f(x)$							

© Cengage Learning Australia Pty Ltd 2013

3 If $f(z) = 3 \times 5^2 x$ find:

a $f(0)$ _____

b $f(-2)$ _____

c $f(1)$ _____

d $f(2)$ _____

e $f(-3)$ _____

f $-f(3)$ _____

4 If $f(x) = a^x$ find:

a $3 \times f(2)$

b $f(4) \div f(2)$

c $f(-2) \times f(7)$

d $\dfrac{24 \times f(5)}{8 \times f(-4)}$

e $f(-2) \times f(5) \times f(3)$

f $4 \times f(-3) \div f(2)$

g $\left[f(3)\right]^4$

h $\left[7 \times f(-4)\right]^2$

© Cengage Learning Australia Pty Ltd 2013

i $\left[\dfrac{f(6)}{f(1)}\right]^3$

j $\left[2 \times f(2)\right]^{-3}$

© Cengage Learning Australia Pty Ltd 2013

Answers

1 $y = 2^x$

$f(x) = e^x$

$y = ab^{\lambda x}$

$f(n) = 1.05x^n$

$y = \dfrac{3}{10^x}$

2 a $f(x) = 2^x$

x	-3	-2	-1	0	1	2	3
$f(x)$	0.125	0.25	0.5	1	2	4	8

b $f(x) = -2^x$

x	-3	-2	-1	0	1	2	3
$f(x)$	-0.125	-0.25	-0.5	-1	-2	-4	-8

c $f(x) = (-2)^x$

x	-3	-2	-1	0	1	2	3
$f(x)$	-0.125	0.25	-0.5	1	-2	4	-8

d $f(x) = 10^x$

x	-3	-2	-1	0	1	2	3
$f(x)$	0.001	0.01	0.1	1	10	100	1000

e $f(x) = 10^{-x}$

x	-3	-2	-1	0	1	2	3
$f(x)$	1000	100	10	1	0.1	0.01	0.001

3 a 3 **b** 0.12 **c** 15

d 75 **e** 0.024 **f** -375

4 a $3a^2$ **b** a^2 **c** a^5

d $3a^9$ **e** a^6 **f** $4a^{-5}$

g a^{12} **h** $49a^{-8}$ **i** a^{15}

j $\dfrac{a^{-6}}{8}$

© Cengage Learning Australia Pty Ltd 2013